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"ENHANCING INNOVATION COMPETENCIES IN DEVELOPING COUNTRIES: THE ROLE OF RESILIENCE AND PERSONALITY TRAITS"

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Table of Content

Introduction 6
Literature Review 9
Hypotheses development 18
Conceptual Model 23
Methodology 25
Research Design 25
Data Collection & Sampling 25
Measurement Scales 26
Data Preparing 27
Data Analysis Techniques 28
Reliability 28
Validity 29
Structural Equation Modelling (SEM) 29
Data Analysis, Results and Discussion 30
Data Analysis 30
Descriptive statistics 30
Measurement Scale Reliability 35
Measurement Scale Validity 35
Correlation Analysis 35
Multicollinearity Test 37
Structural Equation Modelling 37
Results and Findings 38
The analysis of Creativity, Resilience and Personality Traits 38
The analysis of Teamwork, Resilience, Personality Traits 39
The analysis of Initiative, Resilience and Personality Traits 39
The Analysis of Critical Thinking, Resilience and Personality Traits 39
The Analysis of Networking, Resilience and Personality Traits 40
Discussion of Findings 42
Conclusion 44
Implications 45
Limitations of the study 46
Future Research 46
References 47

Appendix A 54
Appendix B 56
Appendix C 58

Table of Figures

Figure 1:Gender distribution of respondents 31
Figure 2: Age distribution of respondents..... 31
Figure 3: Distribution of respondents by academic enrolment 32
Figure 4: Distribution of respondent by their academic programs 33
Figure 5: Distribution respondents by their work experience 33
Figure 6: Distribution respondents' work experience by country..... 34

List of Tables

Table 1: Test of Reliability 26
Table 2: Summary statistics of the constructs (in standardized units) 34
Table 3: Test of Reliability 35
Table 4: Correlation Matrix Table 36
Table 5: Test of multicollinearity..... 37
Table 6: SEM Goodness of fit..... 38
Table 7: Summary of SEM results..... 41

Abstract

Enhancing innovative competences has been considered as important learning goal in higher education due to growing needs for professional skills in the 21st century that makes the life challenging and stressful. To cope with this stress, individual needs to be resilient which can be affected by personality traits. This study aims to determine the relationship between personality traits and innovative competences through the mediating role of resilience. A survey was created, and 521 responses were collected through self-reported questionaries from three developing countries namely, Russia, Pakistan, and Nigeria. The data was analyzed using Structural equation modeling (SEM) and the results indicate that openness to experience, extraversion, agreeableness, and conscientiousness enhanced the resilience and that increases innovative competences including creativity, teamwork, critical thinking, initiative, and networking. Whereas neuroticism decreases the resilience, and which negatively affects the innovative competences. Therefore, we can conclude from the findings of this study that individuals high on openness to experience and conscientiousness are better able to cope with the stress associated with failures because of their rich experience with similar situations and can create the much-desired solutions to problem. Besides, with larger social network, extraverted people can increase their resilience level and ultimately enhances the innovative competence. In contrast, neuroticism personalities are overreactive and emotional instable that might make them less resilient, therefore reduce their level of teamwork, networking, and creativity.

Introduction

The modern world has been craving for the innovation since the industrial revolution and its hunger has never been satisfied rather it has been growing ever since. Almost every aspect of the society strives for this phenomenon. The question is why is this so much important? The foremost relatable answer to this question is that it makes human progress (Pellicer-Sifres et al., 2017) whereby economies of the nation thrive. Innovation makes a significant contribution in the economic growth that overflows between countries, industries, and firms (Thakur-Wernz & Bosse, 2022). It is a buzzword, where every leader knows its importance, but they explain in their own ways. For example, Skillicorn (2016) said, creativity is the ability to generate useful and novel ideas which works as a seed for innovation, but it is just an idea until it's not applied or scaled. So, it is basically a capacity to conceptualize and implement innovative or creative ideas that have a real and meaningful impacts (Adepoju & Nwulu, 2021).

Labor force needs are frequently evolving as globalization continues to influence national and international economies (Foster & Yaoyuneyong, 2016). The growing need for professionals is increasing in today's competitive society of 21st century, therefore, the educational institutions are looking way forward to increase the innovation competence among their students. Because young generation is the future of the economy and they may work for organizations after their studies (Ovbiagbonhia et al., 2019). Indeed, educational system appreciate creative thinking but often don't pay attention to the development of innovation competence which prepares students for the uncertain future (Beghetto & Kaufman, 2014a). Instead, they should help students to develop these skills, they are suppressing it but fortunately, this skill cannot be killed, because it is an inherent and natural aspect of human experience (Beghetto & Kaufman, 2014a). Therefore, the most important thing is the development of the environment that supports and encourage the enhancement of innovation competence among students. Where they have hypothetical situations, updated and relevant curriculum and can learn human experiences based on uncertainty and able to think out of the box (Ovbiagbonhia et al., 2019).

Uncertainty has association with risk and where there is risk, there is element of stress and tough times. There is quote related to the innovation, personality traits and resilience. *"Genius is 1 percent inspiration and 99 percent perspiration"* (Thomas Edison, 1932). This quote can be explained as being initiative with creative idea that is risky is just a starting point, but 99 percent is the continuous struggle and efforts require to accomplish it. In the journey of continuous reconstruction, there can be challenges and hardships throughout the way. Then

resilience plays an important role here to deal with those negative circumstances and challenges. Resilience is basically referred to a positive phenomenon which is an ability to recover from the challenging and stressful situations (Zellars et al., 2011). So, for organizations, stressful situations are the part of their daily routine, they require continuous redesigning in their business process, values and behaviors that ultimately demand perspiration and resilience that comprehensively favor it over innovation. Therefore, this it is very important to find the role of resilience between personality traits and innovation competence.

For the development of this productive environment that can enhance and polish the innovative competence among students through resilience, there should be enough research available to get fruitful results from it. But unfortunately, the most difficult challenge is absence of the abundant research. This gap prevents assessing the factors influence innovation and comprehend the function that of innovation in the development and growth of the economy (Daksa et al., 2018). Individual characteristics affect variously on innovation competence. Therefore, in this study, our focus is to observe the role of resilience on the relationship between personality traits and innovation competence. Although, many studies have been done on innovation competence in relation with psychological and personality traits that focused on the individual level, but they are too limited and narrow (Beghetto & Kaufman, 2014). Hence, in this present study the variable constructs are broad; innovation competence and personality traits constructs each consist of five dimensions

In addition to these three constructs, we have also collected data from three developing countries: Russia, Nigeria, and Pakistan. It is also one of the gaps to be filled in this study because no researcher has studied the relationships between these three variables in these developing countries. We analyze these developing countries because innovation plays a very crucial role in determining their long-term survival, value creation, competitiveness, and productivity (Beghetto & Kaufman, 2014a). Developing countries lack human capital, low managerial capabilities and most important thing is takes time to adopt the technology that is rapidly altering the economies globally. Furthermore, there are many empirical studies on the determinants of innovation in developed countries (Beghetto & Kaufman, 2014a) but to generalize the conclusion drawn from these studies might be inconclusive and misleading for the developing countries (Daksa et al., 2018). So, the main research gap in our study is to understand the role of resilience in enhancing the Innovation competence and personality traits. There is no such study who has used resilience as a mediator variable between Innovation competence and personality traits.

Therefore, the main purpose of this study is to predict the role of resilience between innovation competence and personality traits among students from different study fields in Russia, Nigeria, and Pakistan. The specific objectives are to:

Research Objectives

1. Understand the relationships between innovation competence, resilience, and personality traits among students in three developing regions.
2. Fit a model that predicts individual innovation competence using personality attributes and resilience.
3. Make recommendation to business and educational stakeholders on how to enhance the innovation competence of students, current and future labor force.

Research Question

All this led us to define our research question as:

1. What is the role of resilience in the relationship between big five personality traits and five dimension of innovation competence?

Literature Review

This section will discuss about the previous studies related to innovative competences, personality traits and resilience. Afterwards, we will discuss about the theoretical model and develop the hypotheses accordingly. In the last, we will develop the conceptual model to show the relationship among variables.

Innovative Competences

Innovation is the crucial and important factor for the success of organization and nowadays companies are more focused on creativity and innovation to hire the individuals. Creativity is defined as creation of unique, new, and creative ideas and innovation means successful implementation of those ideas (Farr & West, 1990). Thus, it means before being innovative, students should be creative enough to generate the ideas so later that idea could be successfully converted into a realistic and implementable solution. These factors help to develop the competences. McClelland, (1987) defined the competence is combination of motivation, knowledge, skills, and abilities to increase the performance in any area or aspect of life. Currently, one of the major competences has been considered that is innovation. This competence is the cluster of competences including skills, abilities considered as innovation competences (Marin-Garcia, Andreu-Andrés, et al., 2016).

According to (Farr & West, 1990) innovative competences is expressed as to create the original, relevant, implementable solutions of problems. Developing innovative competences among students has been important among higher education students because in recent years, the work environment is highly challenging due to technological change and complex demand system (Chell & Athayde, 2009a). Besides, universities are being aware to educate the graduates not only being creative but also innovative that can be able to generate the implementable solution of the problems.

Innovative competences have been considered as the important learning goal in higher education (Chan & Yuen, 2014) due to growing needs for professional skills in the 21st century, Besides, innovative competences are considered that can integrate the skills, attitude and knowledge and innovative students are found to have good leadership skills, are motivated towards tasks, creative, self-persistent, take calculated risks and like to work in complex and ambiguous environments that can appear more challenging to them. Therefore, universities are looking for ways that can foster innovative competences among students. Based on Dalke et al., (2007) one of the ways can be a learning environment that is more focused on hands-on

practice and makes the students more engaged in the activities than the traditional learning classrooms and should have supportive teachers. Besides, when goals are clearly expressed and defined and students realize that can help in the development of a career then the students perceive that learning as innovative and that could help to develop the innovative competences (Beghetto & Kaufman, 2014b).

Since decades, innovation competences are the point of discussion for academic institutions and corporations. Therefore, adequate literature is available on this topic and FINCODA (Framework of innovation competences development and assessment project) is one of them. It is most reliable and valid tools available for the measuring and developing the innovation competences (*https://www.fincoda.eu/).* The FINCODA model is devised to assess and increase the innovation capacity of an individual and to fill the gap between academia and workplace. This model is based on five dimensions of innovative competences those are creativity, teamwork, critical thinking, initiative, and networking (Marin-Garcia et al., 2011; Pérez-Peñalver et al., 2018).

Creativity

Creativity is defined as the generation of novel ideas, its improvement, and problem-solving skills (Pérez-Peñalver et al., 2018). It can be a crucial factor that can enhance innovative competences. Because when a problem is identified then there is a need for an innovative solution (Antonietti et al., 2011). Innovative solutions can be developed when students are open minded for new ideas and can see it from different perspectives and become aware of many factors that could affect the decision including possibility and constraints. Besides, it can connect the several events and factors that can be useful to reach a decision making by combining the old and new experiences. Creativity could be like questioning assumptions, leveraging the power of imagination, and challenging the beliefs and stereotypes.

Critical thinking

It is defined as the process after the creation of ideas. It includes evaluating the ideas based on their advantages, disadvantages considering risk in it (Saatci & Ovaci, 2020a). Those students who think critically would like to solve real life ambiguous and complex problems that can be highly challenging for them tend to be more innovative and willing to take more risks (Tabak & Barr, 1999).

Initiative

After evaluation of ideas, there is need to implement those ideas. Individuals who take initiatives or actions are risk takers and they believe in implementation of ideas which enhances the innovative competences. Whereas risk avoidance can make a person less innovative(West, 2002). Risk-taking and confidence are the parts that can make the person innovative and take that idea into successful implementation (Campbell et al., 2004).

Teamwork

Teamwork is defined as to work effectively in groups with collaboration (Pérez-Peñalver et al., 2018) and clear about the goals. When students are clear about their end goal, they can collaborate, mobilize, and work with other followers to achieve that goal that is indication of team-work skills because innovation can never happen in isolation (Chell & Athayde, 2009). Click here to enter text. They will have faith in their skills, abilities, and self-confidence to achieve that required task.

Networking

Networking is explained as to create and nurture the social relationship and build social network. In innovative competences, the process of generating the ideas is not the same as the implementation of ideas because for implementation, students need to consider the several aspects, context, their characteristics, problems that require the critical and innovative thinking and responses that may arise so that students can tackle the difficulty and solve that problem. So, (Marin-Garcia et al., 2011) suggested that networking and communication can be the factor to collect the several constructive ideas to process and assess it through arguments.

To sum up, being innovative and have innovative competences is the process and it cannot be acquired by radical change. Therefore, in the educational system it is unnecessary to strive for radical innovations (Veryzer Jr, 1998) and should believe in process. For instance, in the case of education, students are provided with the open-ended tasks to develop competences in the challenging environment. Another could be like they are working in an innovative setting of a project which builds creativity, need to be social and collaborate with a team to transform the unique ideas into multiple solutions (Oshio et al., 2018). These solutions can bring the value in students and students become more creative that can enhance the innovative competence among students. Therefore, we can say that it is the process which can be learned and developed that can be benefitted from collaboration (Sawyer et al., 2003). Also, it can vary based on context because competences are defined as alignment, integration and adoption of skills,

knowledge, and performance in the specified context (Mulder, 2012). The influence of individual characteristics and context can also seem to be significant in developing innovative competences.

Resilience

To enter in the adulthood from the age of 19-29 years is highly challenging and stressful that includes a lot of ups and down but also open the doors of various possibilities (Arnett, 2015). Further, this phase of life seems to be highly complicated because it includes self-exploration and self-discoveries, and it became more complicated when students enter in university life. The problem they face is like somehow similar at the beginning of years: The environment adaptability, to learn new methods and learn different things, deal with hectic schedule, can easily be frustrated because of less time available. This inflexibility of time leads to lower the performance of students which ultimately decreases the motivation and concentration among students. The individual who finds the ways to cope these challenges are being more resilient and become able to manage and overcome the stress and difficulties in their adult stage. Resilience is defined as the set of personal qualities which enable person to flourish in the adversity (Connor et al., 2003).

The explanation and definitions of resilience has been relatively challenging over the period for academicians because it includes various contexts, namely educational context, community, organizational, clinical, and military. Firstly, this concept was used in physics to define the properties of elastic objects then further it extended for other fields (Dyer & McGuinness, 1996). Generally, the definition of resilience is divided into two group thoughts. The first group defines that resilience is the personality trait and considered as development factor (Lee et al., 2012). The second group defines that resilience is dynamic process and can be influenced by the external factors or environment (Lee et al. 2013). It is process which defines that the ability to adopt well the adversity and stress. Those students have great level of resilience then they can cope with adversity, stress, and difficult situation. While some students are more resilient than others which can leads to lower performance, motivation and even in worst case university with drawl or dropout. Therefore, researchers have found that those individuals who are resilient can engage and adopt and cope with adversity and turn the stressful situation into opportunities which can be useful for learning and development (Campbell-Sills and colleagues, 2006) and can lead to better academically also increases the motivation among students.

Personality traits

One of the most essential themes in psychological research is Personality (Danny Azucar et al. 2017). Research has shown that personality has been demonstrated by the foretelling of many exposures of life, including job performance (Andrew Nealet al. 2011), health (Stephen Soldz & George.E, Vaillant,1999), political attitudes (Alan S.Gerber et al. 2010), academic success (Meera Kommarraju et al. 2009), subjective wellbeing (Natalie Hayes & Stephen Joseph, 2002), Online behaviors (Shaojung Sharon Wang, 2013) and social status (Ameron Anderson et al. 2001). Human Personality is not only a composite of one factor, and this is very subjective in nature therefore it is considered as a complex element of human. According to Sigmund Freud, the first person who gave the theory on personality in 1923. He divided the personality into three parts, the id, ego, and superego and according to Freud's psychoanalytic theory, all these components develop at different stages of human lives like id is the innate element of the personality, whereas Ego develops from the Id and is the decision-making element and Superego is the voice of conscience (moral sense).

The term personality is defined in different ways and many Psychologists have worked on this term for decades and added value addition in this definition. Traits are the attributes that typically belong to a person and have different features. Personality traits are the individual factor that varies from person to person based on the characteristic sets of the behavior, thinking and feelings. It holds the chain of factors, opinions, moods and attitudes that is easily understood while interacting and communicating with other people (Philip S.Holzman, 2022).

More specifically, from human to students, they are the most important part of every nation, and this part is also an important stage of human life. From primary to post graduation or all stages of education, they behave differently, their worldviews are different in various educational and social settings because they have distinguishable personality traits (Soraya hakimi et al. 2020). Understanding the personality traits among students is considered as an important factor, because it clarifies the many insights of students' behaviors in the real world. Personality traits and cognitive abilities are directly or indirectly correlated with their academic achievements (Paunonen & Ashton, 2001). And student's academic achievements lead to many different elements like their motivation, positive attitude towards corporate life and can be valuable resources of any Economy. There are cumulative effects of academic performance and school adjustments with the course of time that have continuity effects on personality (Caspi et al. 2005).

In addition to these definitions, after years of effort, Psychologists have created the five core dimensions of personality traits and became the main kinds of the personality structure (Xiaojie Zhang et. al, 2020). Personality traits are individualistic different characteristics among students, how they think, behave, and feel differently. There are many ways of measuring personality traits but among the best ways to measure those traits are Big 5 personality traits (Preethi CM and R G Priyadarshini, 2018). Personality of a person can be an external or internal factor that is affecting the emotions and it is not only the isolated part of human learning it can be evaluated based on the traits of the society, groups and at certain levels of humankinds (Oguzhan and Sebnem, 2015). In Psychological trait theory, Big 5 personality traits theory was developed in 1949 by D.W. Fiske and it was expanded by the other researchers later onwards and these Big 5 personality traits include Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism.

Openness to experience

Openness to experience is simply how open minded a person is to new ideas or experiences, and it is a tendency to be creative and aesthetically sensitive (Xiaojie Zhang, 2020). It also reflects broad-mindedness of individual, permeable awareness, in depth attitudes (Hakimi, 2011). Openness is further divided into its sub traits like Imagination, Emotionally, Intellect, liberalism, Adventurousness and Artistic interests (Srivastava, 2018). The person who scores high on openness will be categorized as intellectual, aware of his feelings, strong in imagination and thinks in non-confirming ways. On the other hand, if a person scores low on openness they will be categorized as down to earth and do things in a straightforward way (Srivastava, 2018).

Conscientiousness

Conscientiousness is defined as the person extent to which it is responsible and achievement oriented (Jirásek & Sudzina, 2020). This trait is considered as one of the most important traits in personality (Ashton, 2022). When conscientiousness level is high that could lead to positive outcomes in many aspects of lives including student and work life (Ozer & Benet-Martinez, 2006). Due to technological change and emerging new fields, people want to change their personality voluntary that increases the people desires to enhance the conscientiousness (Hudson & Fraley, 2015).

Extraversion

Extraversion is one of the dimensions of personality trait and it is explained as the extent to which person is self-oriented or socially oriented (Feist, 2019a). When individuals have the trait of extraverts that tend to be talkative, cheerful, confident, enjoy spending time with others, good at leadership skills and considered more satisfied from lives (Yao & Li, 2021a). Whereas, when person has low level of extraversion considered as introverts and they are more Quiter, un-social, keep themselves behind the background and enjoy being alone.

Agreeableness

Agreeableness is defined as extent to which person is friendly, considerate and interact with others in friendly manner (Saatci & Ovaci, 2020b). Besides, it is described as having harmonious relationship with others, likeable, kind, warm and pleasant. These individuals are motivated to maintain the positive relationship with others, and they could have good networking and teamwork skills (Graziano & Tobin, 2009). Being agreeableness could have negative impact on the personality because to avoid the conflict they might not be able to think creatively and doubt on themselves and their ideas. The people those are low on agreeableness tend to be consider as unfriendly, self-centered, and uncooperative (Chang et al., 2014) and they share their ideas to other team mates regardless of caring about conflict which can affects the relationship with the teammates.

Neuroticism

Neuroticism is about the extent the person is emotionally stable or emotionally unstable. These neurotic people experience negative feelings and overreact to the situation (Widiger, 2009). When the individuals have high score in neuroticism that means they will experience more negative emotions than the average person like fear, anxiety, guilt, anger and depression (Widiger, 2009). Most of the time, they poorly respond to stress of environment and consider the ordinary situation as threatening that could leads to frustrations. These individuals often tend to be shy, self-conscious, and quite difficult for them to control the urges and impulses which can make them sad or upset (Widiger, 2009).

Theoretical framework

Theory of resilience

Resilience theory described as the stress coping ability that could managed the stress, depression, and anxiety in more effective way. It is the motivational force that drive the human being to pursue wisdom, altruism, and harmony with the force of spiritual strength. According to Benson, (1997) resilience helps to identify the individual traits and strengths that could help the human being to survive in the face the adversity. Besides, Emmy Werner (1982) described the personal characteristics that includes socially responsible, extrovert, conscientiousness, tolerant, adaptable and have good self-esteem. Whereas Norman Garmezy (1991) explained the resilience theory from different perspective and the characteristics he suggested self-discipline, internal locus of control, critical thinking skills, self-esteem, and problem-solving skills. There are many resilient qualities describes by authors included creativity (Simonton, 2000). These list of traits, states and characteristics has been exhaustive therefore the paradigm shift from identification of risk factors to focus on personal strength and qualities has been significant.

Resiliency theory explains that there is force inside everyone which drive them to seek wisdom, self-actualization, harmony, and altruism with the help of spiritual strength. In educational system, resilience helps the students to grow and know themselves by body, mind and spirit through innate qualities or strength. This helped the students to handle the adverse situation without being overwhelmed. Besides, it produces the child-like nature among individuals and create the self-love that drives them to enhance the creativity, being adventurous and curious in life. Resilience can be strengthened by the personal qualities or personality traits which can help the students to become more innovative and creative. Whereas (Rutter, 2013) explained the resilience theory consider the individual difference in personality traits; genetics creates the difference in the response of each person towards the risk and protective factors because same person can respond differently in different situation, so it is more affected by environment change. Similarly, (Luthar, 2013) align with his point of view and he explained that resilience is dynamic process that is related to positive adaption of environment in case of adversity. Therefore, we can say that resilience is affected by personality traits and by the environment which can affects the students' innovative competences in-terms of creativity and critical thinking skills.

Investment theory of creativity

This study has been used the basis of investment theory of creativity proposed by Sternberg (2006) with the collaboration of Todd Lubert. He proposed that creativity involves largest part of decision. People are considered creative when they buy low and sell high in terms of ideas. Buy low means creative people propose the ideas which are novel and can get high level of resistance. But they convince people, and their ideas start getting acceptance, so those individual sell at high and reap out the profits then again move to another new idea. These creative people resist to follow the crowd instead they create their own direction and try to achieve it. Creativity is learned behavior not inborn behavior, they set their own attitudes and work in that direction by taking the risks, face the obstacles and examine their own thoughts which can make the hurdle in the creative process. Based on investment theory of creativity, these set of attributes are teachable and students can adopt these attributes and enhance their creativity.

Investment theory of creativity is explained based on six components those are: Abilities, Style of thinking, environment, motivation, knowledge, and personality traits. *Abilities* includes to see the problems from new perspective and try to solve it with new ideas and then evaluate those all ideas and try to pursue the feasible one. Afterwards, try to convince others the value of that idea and take their support and eventually sell that idea to those people. Second component is *style of thinking*, that creative person is more open minded and open to new ideas and think in a way that can valuable and fit for its own purpose. Third component is *environment* that creative people need supportive environment to flourish their ideas, therefore, in universities teachers should provide supportive environment to students to enhance the creativity. Besides, creative people love to do what they do that is their main cause of *motivation* and that enhance the creativity. Creativity cannot be totally based on novel ideas rather it is also affected by existing *knowledge* and new information. Finally, *personality traits* play a very important role in the creativity, these individuals are risk seekers, and they love to take risk and they do not try to do the things through conventional way rather they buy low and sell at high and show the readiness to take the sensible risks and challenges and overcome the obstacles and have a sense of clarity for their ideas. Overall, these six components can be adopted which can make the individual creative and innovative.

Hypotheses development

Personality traits and innovative competences with the mediating role of resilience

Resilience is considered as process that can demonstrate itself by the response of external life events and personality profiles of individual (Dyrbye & Shanafelt, 2012). Besides, resilience is defined as the personality characteristics that can reduce the negative effects of stress and help in adoption. The relationship between resilience and personality traits has been discussed by various authors that provides higher level of conformity. It has been investigated on university students and found significant relationship between resilience and personality traits including extraversion, neuroticism, conscientiousness, agreeableness, and openness to experience (Campbell-Sills et al., 2006). Moreover, resilience is not only explained by fixed personality or personal characteristics rather it can be dynamic developmental process that reflects the positive adoption even in the case of severe adversity (Luthar, 2003). She explained that it involves the process and conditions that interacts in a way to deal with adversity that determines the resilience which can affects the innovation competences of students.

Openness to experience and innovative competences with the mediating role of resilience

Openness to experience is the one of the main components of personality traits. It is defined as intelligent, smart, broad minded, and independent. Individuals with high level of this characteristic have wide varieties of interests and open to different experiences. This personality trait is getting attention from a lot of researchers because of its effects on physical and mental health of students or individuals. Besides, (Lee-Baggley, Preece, & DeLongis, 2005) suggested that openness to experience individual tends to be more adaptive and can manage the stress effectively which shows the high level of resilience.

Openness to experience individuals find the challenges and novelty as rewarding than threatening which enhances the resilience level in individuals (Williams, et al, 2009). Those students who are open to experience are more knowledgeable and intellectual because they are open to new ideas. Simultaneously, their divergent thinking enables them to open to different ideas and understand the problem considering various perspectives (Yao & Li, 2022) that encourages the creativity. Moreover, Beaty et al., (2016) & Feist, (2019) found the positive relationship between innovative competences and personality traits and suggested that openness to experience encourages the creativity in people that is component of innovative competences. We can infer that openness to experience positively affects the innovative

competences, So, we can say that openness to experience enhances the resilience and resilience enhances the creativity.

After the creation of ideas, those ideas need to evaluate and think in critical way. Critical thinking is about generating ideas and evaluates them based on their risks, advantages, and disadvantages (Saatci & Ovaci, 2020). These individuals are open to ideas and more open-minded (Sahanowas, 2021) suggested positive significant relationship between resilience and critical thinking by doing the studies on first-year undergraduates by collecting the data from 490 participates from west Bengal. We can infer that those people are more open to experience tend to allow the ideas and can have high level of resilience and can take the sensible risks and evaluate them which enhances the critical thinking. Those individuals who are risk takers tends to take more initiatives because they are resilient enough to bear the risks and bring the novel ideas. When people are open to new ideas, they can build good social network which enhances the resilience among individuals and can be able to work in teams (Tawfik et al, 2017). Therefore, we can conclude that openness to experience enhances the resilience and resilience encourages the innovative competences among people. Here resilience plays as mediating role, so we developed the following hypothesis and sub-hypotheses.

H1: Resilience plays a mediating role between openness to experience and innovation competences

H1a: Resilience will mediate the positive effect of openness to experience on creativity

H1b: Resilience will mediate the positive effect of openness to experience on initiative

H1c: Resilience will mediate the positive effect of openness to experience on critical thinking

H1d: Resilience will mediate the positive effect of openness to experience on Teamwork

H1e: Resilience will mediate the positive effect of openness to experience on networking

Neuroticism and innovative competences with the mediating role of resilience

Neuroticism is defined as when person is unstable and overreact to situation and experience negative feelings. Neurotic people might not try to understand the situation and try to solve it calmly, rather make the situation more miserable which can show the lower level of resilience. Nakaya, Oshio and Kaneko (2006) determined the relationship between personality traits and resilience using 130 teenagers' sample and found that neuroticism negatively affects the resilience, so the person with high resilience will not show the neurotic tendencies like

depression or stress. Likewise, (Froutan, et al, 2017) suggested that neuroticism has significant relationship with resilience that person with low neuroticism will perceive lower level of stress and adversity and that person can be less resilient.

Moreover, neurotic individuals can suffer from anxiety, have impulsive thoughts and behavior and anger towards others that shows the lower ability to manage the stress and anxiety and become less resilient. This emotionally unstable behavior can hinder the level of creativity (Ali, 2019) and that is quite unfavorable for innovation. These individuals might not be able to evaluate the ideas or handle the risks or take initiatives due to lower level of resilience which negatively affects the critical thinking. Emotionally unstable people find it difficult to work in teams and cope with that stress because they overreact in every situation which negatively affects the teamwork and build network. Therefore, we can infer that neuroticism decreases the resilience and due to lower resilience, that leads to lower innovative competences. So, we developed the following hypothesis and sub-hypotheses.

H2: Resilience plays a mediating role between neuroticism and innovation competences

H2a: Resilience will mediate the negative effect of neuroticism creativity

H2b: Resilience will mediate the negative effect of neuroticism initiative

H2c: Resilience will mediate the negative effect of neuroticism critical thinking

H2d: Resilience will mediate the negative effect of neuroticism on teamwork

H2e: Resilience will mediate the negative effect of neuroticism on networking

Extraversion and innovative competences with the mediating role of resilience

Extraversion is defined as people who are energetic, expressive, active, gregarious (Yao & Li, 2021) social, confident, and excitement seeker (Feist, 2019). Extrovert individuals love to socialize with other individuals and get the peer support which can help them to cope and manage the stress effectively which enhances the resilience. Besides, (Froutan, et al, 2017 & Shafiezadeh, 2012) found that extraversion positively affects the resilience. Also, this personality trait can bring more positive emotion (Vollrath, 2001) and help to make warm and cordial relationship with others which makes the person more resilient. Due to high level of resilience, extrovert individuals tend to take more risks because they can manage and cope with those risks. This behavior enhances the creativity and critical thinking among people. This risk-taking behavior encourages the initiatives as it stimulates the divergent thinking to bring and

generate new ideas as extravert people likely to share their ideas (Kandler et al.,2016) and open to get the feedback to improve it more. It shows that they can work more collaboratively in teams (Tawfik et al, 2017) and can make the strong social network (Wilks & Spivey, 2010 & Kalaitzaki et al, 2020). Therefore, we can infer that extraversion enhances the resilience and resilience increases the innovative competences and based on that we developed the following hypothesis and sub-hypotheses.

H3: Resilience plays a mediating role between extraversion and innovation competences

H3a: Resilience will mediate the positive effect of extraversion on creativity

H3b: Resilience will mediate the positive effect of extraversion on initiative

H3c: Resilience will mediate the positive effect of extraversion o critical thinking

H3d: Resilience will mediate the positive effect of extraversion on teamwork

H3e: Resilience will mediate the positive effect of extraversion on networking

Conscientiousness and innovative competences with the mediating role of resilience

Conscientiousness is defined as the extent to which an individual is persistent, determined, consistent and focused to achieve their goals (Jirásek & Sudzina, 2020). These individuals are achievement oriented (Ali, 2019) and they prefer planned actions over spontaneous actions or behavior (Stock et al., 2016) which shows the higher level of resilience. This trait can help to make better decisions and organize well, plan and think before taking any action. So, people with conscientiousness consider the problem as challenge or problem-solving issue with shows the high resilience. Besides, Shafiezadeh, (2012) found that conscientiousness positively affects the resilience. When people are highly resilient, focused, persistent, and competitive those can achieve the high performance in many aspects of lives (Ali, 2019) can enhance the creativity.

Previous studies have considered creativity as traits which involves, adoptability, initiative, flexibility, originality, spontaneity, and ingenuity (Barron, 1969) and all these traits are related to resilience and problem-solving skills which determines the connection between creativity and resilience. Besides, Thomson, (2008) empirically found the significant relationship between creativity and resilience. De Caroli and Sagone, (2014) found among Italian adults that there is significant association in some aspects of creativity, adaptability and competences and it shows that when they perceive themselves more creative and they are more

motivated to find new solutions in a resilient way. Therefore, we can say that conscientiousness enhances the resilience and resilience enhances the creativity.

When individual is creative which brings a lot of novel ideas and to evaluate those ideas based on existing risks they need to think more critically. Critical thinkers are well-informed, curious, truth seeker, persistent, open to criticism, open to personal bias and open-minded (Facione, 1990). Furthermore, (Reyna del Carmen Martínez Rodríguez) explained that critical thinking allows students to reflect their ideas based on reality despite having all the pressure from the environment and they should be free to express their feelings that could increases the problem-solving ability among students which can be due to higher resilience. When individual is determined, focused and persistent then it can enhance the resilience and due to high resilience, they tend to be more critical thinker. Due to open to more ideas and high-risk taking behavior they would be take more initiatives and could face the challenges by the social support and supportive network. (Tawfik et al, 2017) found that those people are resilient can work more effectively in teams. Therefore, we can infer that conscientiousness affects the resilience then resilience affects the innovative competences, so we developed the following hypothesis and sub-hypotheses.

H4: Resilience plays a mediating role between conscientiousness and innovation competences

H4a: Resilience will mediate the positive effect of conscientiousness on creativity

H4b: Resilience will mediate the positive effect of conscientiousness on initiative

H4c: Resilience will mediate the positive effect of conscientiousness on critical thinking

H4d: Resilience will mediate the positive effect of conscientiousness on teamwork

H4e: Resilience will mediate the positive effect of conscientiousness on networking

Agreeableness and innovative competences with the mediating role of resilience

Agreeableness is defined as people who are caring, trustworthy, cooperative, gullible, forgiving, respectful (Saatci & Ovaci, 2020) and can sacrifice their own opinion in conflicting situation (Chang et al., 2014). They are more soft-hearted, compliant, tolerant (Saatci & Ovaci, 2020) and they prefer the cooperation and positive social and interpersonal relationship. People who show high agreeableness tend to have less stress which means they are more resilient and have good social relationship to avoid the social conflict (Kim et al., 2016). These individuals can perform well in teamwork. High agreeableness increases the level of resilience because it

brings the positive emotions and these emotions made the individuals to build lasting personal resources like social resources, physical and intellectual and it may help the individual to cope with the stress and gain a lot of social support and confidence to deal with situation (Gong, et al., 2020). Therefore, we can say that agreeableness enhances the resilience and resilience encourages the teamwork and social support or network.

Whereas, when individuals have high agreeableness, they wanted to avoid the social conflict which can happen due to disagreements and that could discourage the innovative ideas (Feist, 2019) and ultimately negatively affect the creativity. Nevertheless, creative people challenge the status quo and bring creative solutions that can bring a lot of criticism and resistance but due to high agreeableness they tend to be less creative. Besides, due to less creativity, their critical thinking is also get affected because they would not evaluate the ideas based on its advantages and disadvantages (Saatci & Ovaci, 2020). Sahanowas (2021) suggested that students with critical thinking have divergent views and can be able to show the self-confidence and able to solve various issues in their surroundings and can communicate confidently and open minded. When individuals have high agreeableness, they tend to be less critical because they might not have divergent views and communicate it confidently and tend to take less risk. Risk averse individuals would not be able to take initiatives because they are not confident in their abilities and creative which negatively affects the innovative competences. Hence, we developed the following hypothesis and sub hypotheses:

H5: Resilience plays a mediating role between agreeableness and innovation competences

H5a: Resilience will mediate the positive effect of agreeableness on creativity.

H5b: Resilience will mediate the positive effect of agreeableness on initiative

H5c: Resilience will mediate the positive effect of agreeableness on critical thinking

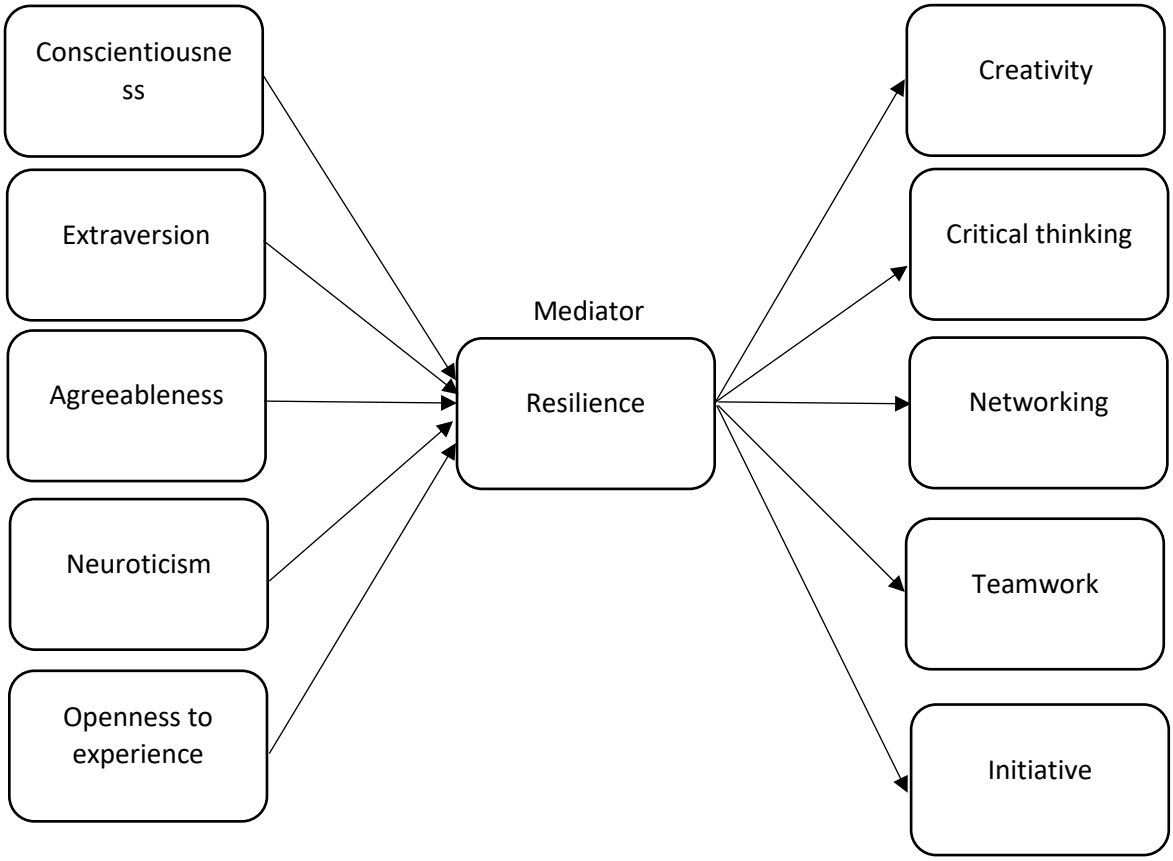
H5d: Resilience will mediate the positive effect of agreeableness on teamwork

H5e: Resilience will mediate the positive effect of agreeableness on networking

Conceptual Model

Based on above theories and hypotheses, we developed the conceptual model. This model explains the link between two variables personality traits (independent variable) and innovative competences (dependent variable) with role of resilience as a mediator on the enhancement of innovative competences. Theory of creativity explained that personality traits enhance the

creativity (Sternberg, 2006) which ultimate increases the innovative competences. Besides, theory of resilience (Benson, 1997) explained that resilience can be affected by personality traits and resilience can be learnt from the environment (Luthar, 2013) so that can help to enhance the innovative competences among students. Therefore, when person possess the personality traits namely, conscientiousness, extrovert, agreeable and open to experience that can enhance the resilience whereas when the person is emotionally unstable and neurotic that could decrease the ability to cope with the adversity. When the person is resilient then it might enhance the creativity, critical thinking, initiatives, teamwork and networking abilities in the students and can perform and manage well in the practical lives (Sahanowas, 2021; Saatci & Ovaci, 2020 & Tawfik et al, 2017).



Methodology

In this section we discussed about the research philosophy, research approach, methodological choice, strategy, time horizon, techniques of data collection, construct of questionnaire, measurement scale for big five personality traits and Innovation competences. Apart from these all, reliability and validity of the data was explained. Finally, explanations and justifications of the data analysis techniques were provided.

Research Design

Research philosophy is a way to inquire the reality or struggle to see the truth through various patterns (Winit-Watjana, 2016). To answer the research questions through selecting an appropriate methodology, this research is following the positivism philosophy. Primary goal of positivism paradigm is to generate a casual or explanatory connection that will ultimately anticipate the aspect of the questions (Park et al., 2020). Further, Hypothetico-deductive model of science was aligned with positivism philosophy, and it will work as a lens that filter out the research one by one. First its start will the theory that is the part of the literature review, second building the hypothesis for testing, third is designing an operation through operationalizing the variable and fourth is conducting an empirical study (Park et al., 2020). So basically, this sums up the deductive research type.

This study adopted a deductive research approach and a quantitative methodological choice. In this choice, we measured the impact of big five personality traits on innovative competences and tested the hypothesis. Data type is primary data and strategy is the survey.

Data Collection & Sampling

Data was collected through electronic survey that was created on google forms and distributed to the students through various platforms. Three countries were considered for this study these are Nigeria, Pakistan, and Russia. All participants were the students and from different fields having qualifications of under-gradation, graduation, post-graduation, and certified degrees. Majority of the data were collected through social media networks to get responses from the participants from 3 countries like, WhatsApp broadcast groups, VK platform, Facebook messenger and Telegram. There were also the other ways to approach the students like professors asked from their students to fill the questionnaire during classes and we approached the students during break time of university by describing the research purpose to fill the questionnaire.

Total 521 questionnaires were collected and out of those 213, 140 and 168 were from Nigeria, Pakistan, and Russia respectively. Time horizon of the research data was cross sectional and there were not any time constraints for this study as it was supposed to be conducted in 5 to 6 months but there was no any variable that needed to be observed for different intervals of the time. English language was used for the Nigeria and Pakistan and whereas for Russian students, separate questionnaire was developed in Russian language.

Measurement Scales

Questionnaire was consisted of four sections, first section is based on demographics and other insights related to paid and unpaid jobs, specialization, study year of the students etc. In second section 34 questions were added for innovative competences, 10 questions for the big five personality traits and 3 questions were added for Resilience variable. All the questions were taken, and constructs were created based on the past papers and theories.

Innovation Competences

Innovation competences is our dependent variable and there are five innovation competences that are used to determine this dependent variable. FINCODA (Framework for Innovation competences Development and Assessment) tool is used for Innovation competences (Marin-Garcia, Andreu Andres, et al., 2016). Further this FINCODA barometer has an adequate validity and reliability (Butter and van Beest, 2017). Innovative competences scale that is consisted of 34-items were used to ask participants to rate the level of teamwork for each item on 1 (absolutely disagree) to 5 (absolutely agree). Just like teamwork, there are total 5 scales on which innovation competences are measured and for each scale there are 6 to 9 items for each scale. For reliability of the scale, Cronbach's alpha is used and is mentioned in the table below.

Table 1: Test of Reliability

Constructs	Alpha Cronbach
Creativity	0.91
Teamwork	0.85
Initiative	0.86
Critical Think.	0.79
Networking	0.80

Big Five Personality Traits

Big five personality trait is an independent variable and there are also five constructs that are used to define this independent variable based on empirical research. There are a lot of research have been conducted on the big five personality traits in the history and there are many versions of scales are available in those studies. Some have used 100-item questionnaire where some have used 60-item questionnaire with 5/7- or 9-point scale (Goldberg, 1992; Hayes & Joseph, 2003). In this research, short version of questionnaire was used with 10-item and 5-point scale ordered from absolutely disagree to absolutely agree (Gerber et al., 2010; Gosling et al., 2003). Each personality trait is consisted of two items including one reversed item. To increase the response rate and reduce the size of the questionnaire, single item measure is used for the resilience that how well this single item captures key characteristics of the construct being examined and how useful it is to practitioners (Fisher et al., 2016).

Resilience

Resilience is used as a mediator variable between innovation competences and big five personality traits which is our dependent and independent variable. There is no previous research on resilience where it is used as a mediator between these mentioned variables at same times. Resilience is measured with the Brief Resilience Scale (BRISC). There are several versions of BRISC (Catalano et al., 2011; Dodd et al., 2015; Shin et al., 2012) and in this research 3-item scale is used which is almost a short version of CD-RISC (Charoensap-Kelly et al., 2021). The 5-point Likert scale is used to measure the resilience ranging from (1) is *absolutely disagree* and (5) is *absolutely agree.* For the reliability, Cronbach's alpha is measured, and it is 0.74 which means scale is reliable to use.

Demographic variables

In addition to Personality traits, Innovative competences, and Resilience, we included the specifications measures of gender (male & female), age (In years), education (including graduation, masters, post-graduation, and certified holders), work experience and finally main variable countries (Nigeria, Pakistan, and Russia).

Data Preparing

Data was consisted of 521 total observations. Further for the final analysis we took three variables for this research. This variable selection was based on the previous work done by other researchers and research gap that were not discussed before and focus on the how better

can help the society on the development of innovation competence of the students for the betterment of their career and future generation.

Pre-processing the data is one of the important steps that makes analysis more useful, and this step has further sub-steps and techniques that are commonly used by researchers globally. First, in our data there were no missing values, but data was consisted of few outliers and to deal with outliers there are several techniques and deleting or removing of the data is one of them. Deleting the data is not considered as a good practice (Church, 2018). Therefore, we didn't remove the outliers as in resultantly it would reduce the observations as well and it is better to use as much as data to generalize the research and because of that it is not one of our limitations of this study. So, rather than deleting the data, we used winsorized them. "*Winsorization is a procedure that limits how extreme a return is allowed to be, with this procedure extremely high and extremely low values in the data set are set equal to specified cut-offs*" (Cao et al., 2018, p.16). We also relabeled the variables and items that are used for constructs to make them more readable also converted the type as a categorical variable.

Data Analysis Techniques

To answer the questions through analyzing the data, first we measured the reliability of scales of innovative competences and resilience with the help of using Stata Software. For reliability, Cronbach's alpha was used, for validity confirmatory factor analysis (CFA) was used and for personality traits…. was used to measure the accuracy and consistency of the constructs. Further, to analyze the relationships Structural equation model (SEM) was explained.

Reliability

Those people are considered as reliable whose behavior can be predictable or can be relied on consistency. Likewise, this concept is also followed in the research field, the extent to which the instrument used in the research, yields the same results on repeated measure will show the reliability of the research (Putka & Sackett, 2010). It is not possible to give the exact figures to measure the reliability but there are three estimations that are commonly used for to calculate the reliability. The three attributes of estimating the reliability are Homogeneity, Stability and Equivalence and there are several ways to measure each attribute (Heale & Twycross, 2015)

The nature of this study asks to consider the Homogeneity attribute which means the internal consistency among the various items that together represent the one construct in the research. In our research, both independent and dependent variables are composed of many items and each item is composed of various questions that are explained earlier. So,

homogeneity means the extent to which various items reasonably team together to measure a single characteristic. Like agreeableness is one of the constructs of the big five personality traits and to determine this we have two items “is generally trusting” and “tends to find fault with others”.

Cronbach’s α is one of the most used techniques used in to measure the homogeneity and ultimately it represents the how reliable the instrument is. In this study Cronbach’s α is used to test the internal consistency among innovative competences constructs and resilience as well. It tells the correlation in every combination among the set of items. It is measured between 0 and 1. The acceptable score for the Cronbach's α is 0.7 or higher than this, which means that instrument is reliable to research further (Heale & Twycross, 2015). Reliability for Innovation competence and Resilience has been tested and that is more than 0.7. This means that scales are valid.

Validity

Validity describes how well the data gathered corresponds to the investigation’s actual area (Ghauri & Gronhuag, 2005). It is basically used to check the accuracy of the data. There are four main types of the validity namely, content validity, construct validity, criterion validity and face validity and each type has further sub-type (Heale & Twycross, 2015). In this study, validity is required to check because it describes that how well the concept, idea or behavior transformed or translated into operating reality and functioning.

Further, Confirmatory factor Analysis method is a technique that is used in this study to check the validity. It is a type of structural equation model that deals with the relationships between factor and indicator (Moore, 2012).

Structural Equation Modelling (SEM)

To identify the effect on innovation competence through personality traits and resilience, the structural equation model is used. The normality of all variables was checked before proceeding with Structural Equation modelling (SEM). Basically, it is a framework of different multivariate statistical method and consist of different domains like econometrics, statistics psychology, and biology. It includes correlation/regression, simultaneous equation, factor analysis, path analysis and measurement theory. This model is used to test the relationships between latent and observed variables. Latent variables are the variables that cannot be observed directly or unmeasured variables and these are measured by connecting to the

observed variables. Observed variables are the variables that can be measured (Beran & Violato, 2010).

Furthermore, SEM is the combination of two components: measurement model and structural model. In measurement model, confirmatory factor analysis (CFA) is used based on SEM. One of the main advantages of CFA is it helps in bridging the often-observed gap between theory and observation (R.O. Mueller., 2001). CFA deals with the latent variables, "*it extracts the latent constructs from other variables and share the variance with related variable*s" (Fan et al., 2016). On the other hand, Path Analysis is very useful tool for testing the hypothesis with both casual effects direct and indirect. Mediation is one of the common functions of the path analysis, where variables affect directly or indirectly to the outcome through other variables (Fan et al., n.d.). Like in this study, Resilience is playing a mediation role and we want to see indirect effect of personality traits on innovation competence through it.

Data Analysis, Results and Discussion

In this section, first, we provide the results of the different statistics analyses ranging from descriptive, to correlation and to structural equal modeling. In addition, we also report and discuss several other diagnostics statistics. Second, we discuss the findings of the structural equation modelling

Data Analysis

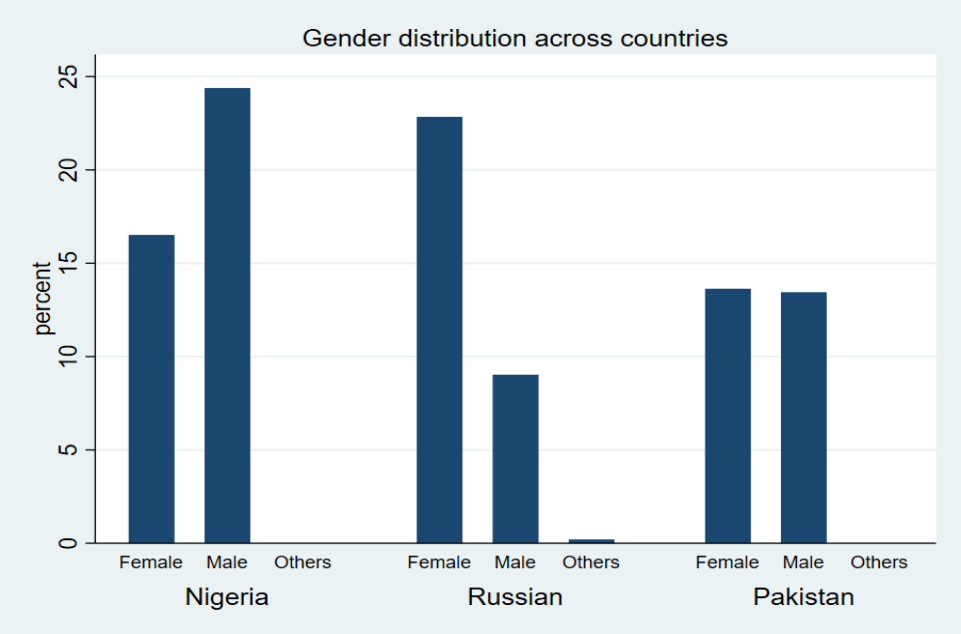
Descriptive statistics

Several demographics variables were collated from the respondents like country of origin, gender, age, work experience and academic program enrolled in. These variables were analyzed to understand the sample structure and characteristics.

Gender distribution

More female students participated in the survey than male, accounting for 53% and 47% respectively. The insignificant balance (0.19%) represents the non-conventional sex category which is not important for our study. Furthermore, the gender distribution by country revealed that unlike Pakistan with equal number of male and female, female participation is greater in Russia than in Nigeria. See Figure 1 for details.

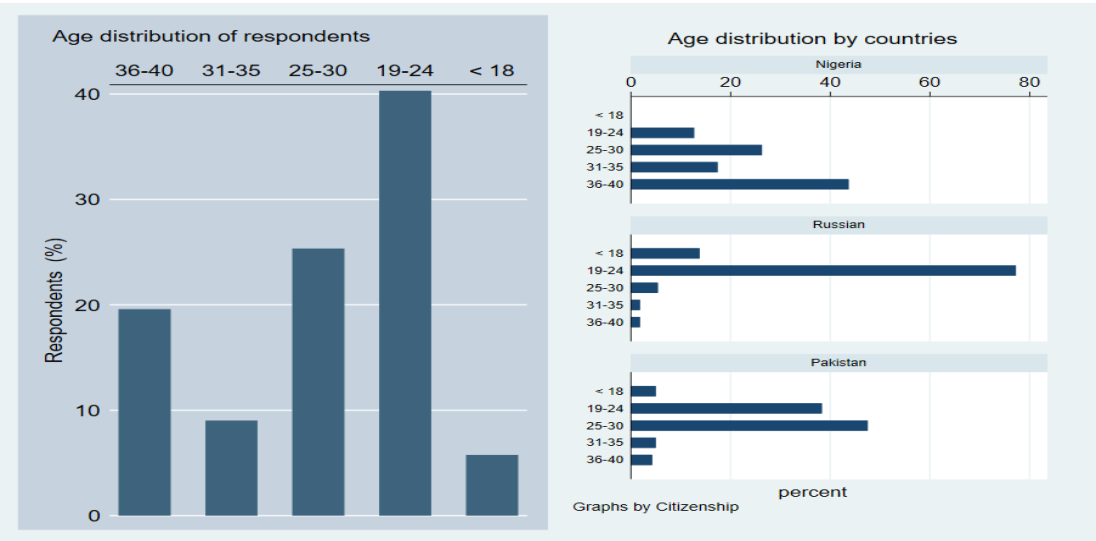
Figure 1:Gender distribution of respondents



Age distribution

About 65% of the respondents are between the ages of 19 and 30, another 19% between the ages of 36-40. In the first age category are students pursuing undergraduates and graduate studies and in the second are doctoral and other post graduate study students. Comparing countries, in general, Nigeria has older students with 50% of them over 30 years while more than 60% of the students in Russia and Pakistan are below 30 years old. Figure 2 provides more details.

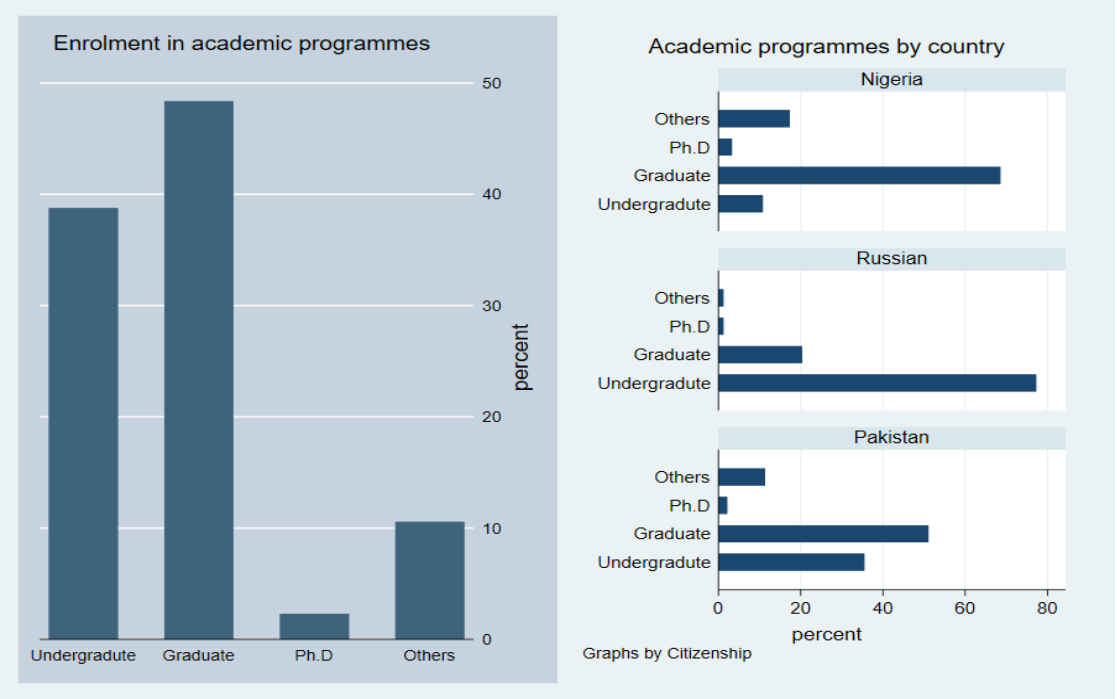
Figure 2: Age distribution of respondents



Enrolments in Academic program

The respondents are at the different levels of their educational pursuits; undergraduate, graduate, doctorate, and other post-graduate programs. Nearly 86% of the respondents are enrolled in either in undergraduate or graduate programs. In addition to these programs, it is common to see students in Pakistan and Nigeria pursuing other professional certifications. Besides, these two regions have higher enrolment for both graduate and doctoral students than Russia. Figure 3 describes the respondents' enrolment.

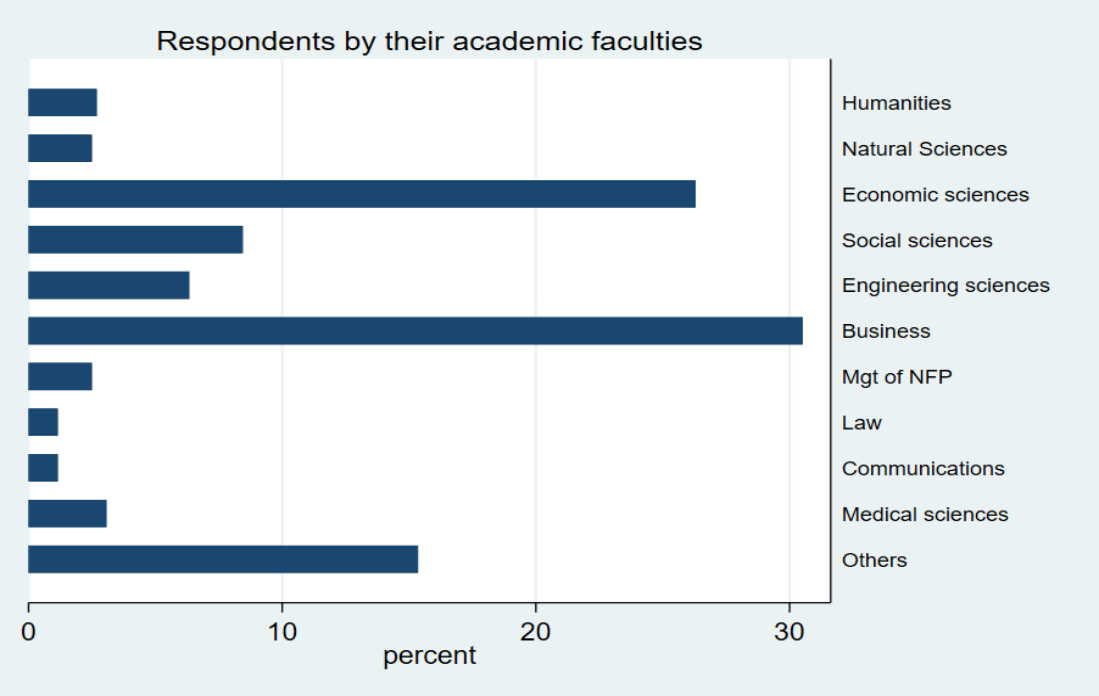
Figure 3: Distribution of respondents by academic enrolment



Distribution of respondents by their academic faculties

Students of the Business, Management, Economic and Social science faculties represent about 70% of the total respondents while medical, natural, and engineering about 12%. Students from other faculties represent 15% of the total sample.

Figure 4: Distribution of respondent by their academic programs



Respondent's work experience

More students (about 70%) have some form (part-time and full-time) of work experience than not. Specifically, about 47% of them have worked for more than 6 months compared with 20% who have no form work experience. Nigeria has the highest number of experienced students and Russia the highest number with no experience. This is the case perhaps because the later has older students.

Figure 5: Distribution respondents by their work experience

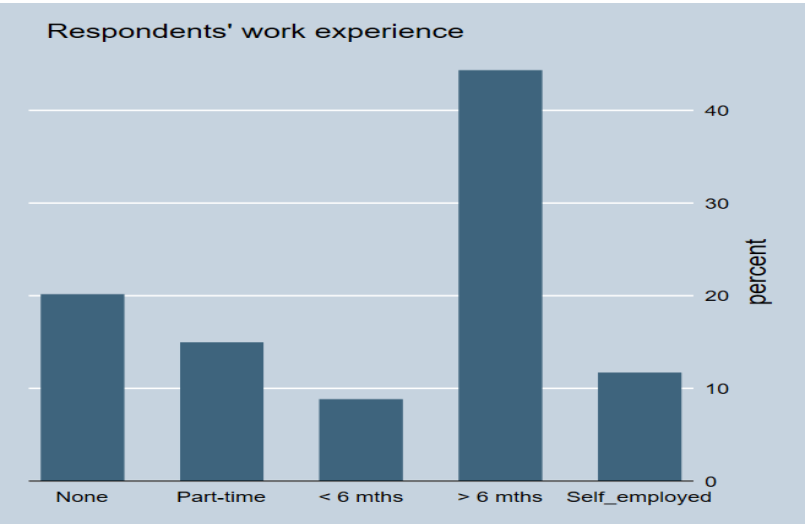


Figure 6: Distribution respondents' work experience by country

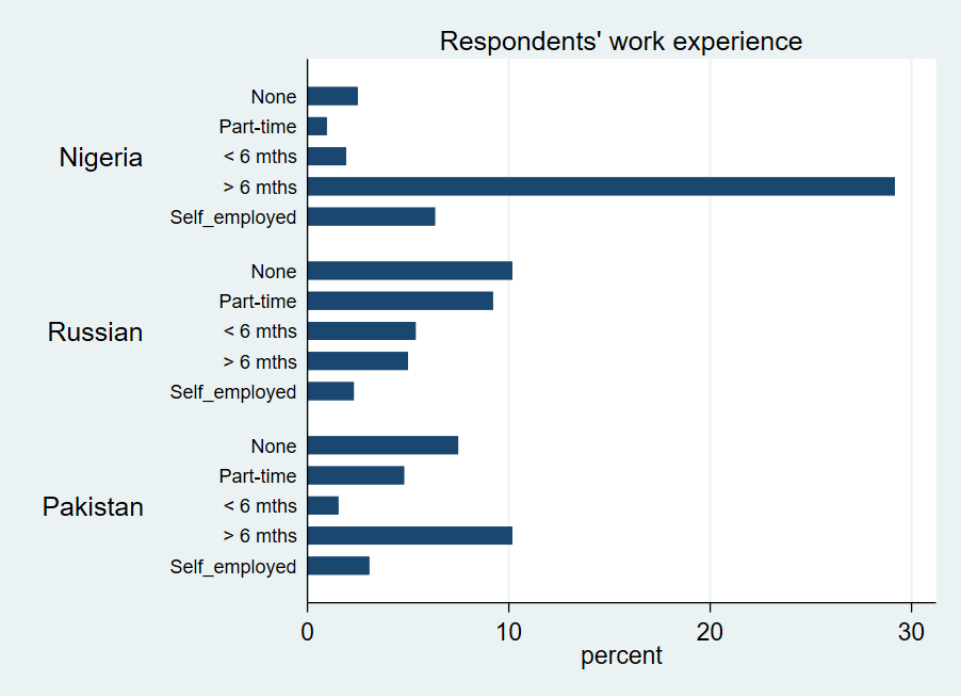


Table 2: Summary statistics of the constructs (in standardized units)

Variable	Mean	SD	Skewness	Kurtosis	Min	Max
Agreeableness	0.028	0.929	-0.655	2.475	-1.930	0.999
Extraversion	0.047	0.904	-0.282	1.939	-1.470	1.171
Conscientiousness	0.030	0.928	-0.480	2.263	-1.825	1.092
Openness	0.100	0.790	-0.311	1.732	-1.099	0.943
Creativity	0.035	0.898	-0.582	2.779	-2.110	1.322
Teamwork	0.043	0.873	-0.507	2.522	-1.984	1.260
Initiative	0.033	0.908	-0.539	2.640	-2.060	1.308
Networking	0.035	0.904	-0.401	2.469	-1.895	1.421
Critical think	0.030	0.927	-0.131	2.372	-1.828	1.669
Resilience	0.018	0.959	-0.253	2.042	-1.808	1.259
Neuroticism	0.016	1.000	0.077	1.966	-1.550	1.586

Measurement Scale Reliability

As part of the study procedures, we tested all the measurement scales for reliability. A scale is reliable if measures the unobserved construct consistently. Specifically, we adopted the internal consistency method of reliability using Alpha Cronbach. In line with the acceptable convention, the Cronbach Alpha for all constructs is above the minimum acceptable threshold of 0.70. The Alpha Cronbach for the constructs range from 0.74 to 0.91.

Table 3: Test of Reliability

Variables	No of items	Alpha Cron.	AIC
Creativity	9	0.91	0.53
Teamwork	7	0.85	0.46
Initiative	6	0.86	0.52
Critical Think.	6	0.79	0.38
Networking	6	0.80	0.41
Resilience	2	0.74	0.59

Measurement Scale Validity

As part of the analysis procedure, SEM measurement model was used to model the unobserved latent constructs of resilience, innovation competences and obtained satisfactory model fit indices. Specifically, we report the following goodness of fits Comparative Fit Index (CFI) = 0.9, Tucker Lewis Index (TLI) = 0.9 and RMSEA = 0.06.

Correlation Analysis

Correlation co-efficient measures the strength and direction of the variation in one variable associated with changes in the other variables. The correlation analysis result for this study is reported in Table 4. The results show that all dimensions of personality traits, except Neuroticism, have a significant and weak positive correlation with creativity, teamwork, critical thinking, networking, initiative, and resilience. The pair-wise correlation co-efficient (r) among these variables range from 0.24 to 0.41. On the other hand, a no significant relationship is reported with between neuroticism and all dimensions of innovation competences.

Table 4: Correlation Matrix Table

	1	2	3	4	5	6	7	8	9	10	11
1. Agreeableness	1.000										
2. Extraversion	0.216***	1.000									
3. Conscientiousness	0.232***	0.146**	1.000								
4. Openness	0.237***	0.192***	0.243***	1.000							
5. Neuroticism	0.101*	-0.115*	-0.013	0.102*	1.000						
6. Creativity	0.330***	0.326***	0.384***	0.410***	-0.088	1.000					
7. Teamwork	0.317***	0.275***	0.437***	0.378***	-0.078	0.709***	1.000				
8. Initiatives	0.296***	0.313***	0.410***	0.371***	-0.080	0.803***	0.674***	1.000			
9. Networking	0.317***	0.337***	0.355***	0.339***	-0.089	0.747***	0.720***	0.709***	1.000		
10.Critical Thinking	0.287***	0.239***	0.273***	0.337***	-0.010	0.768***	0.619***	0.677***	0.678***	1.000	
11. Resilience	0.248***	0.275***	0.383***	0.309***	-0.0986*	0.445***	0.407***	0.442***	0.400***	0.310***	1.000

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Multicollinearity Test

Multi-collinearity occurs where two or more independent variables have high correlations among themselves. This might make the path co-efficient spurious. The presence of multicollinearity can be detected using the variance inflation factor (VIF) statistics and a VIF more than 10 indicates a multicollinearity problem. The test result in Table 5 reports a satisfactory VIF values for all the independent variables.

Table 5: Test of multicollinearity

Variable	VIF	1/VIF
Resilience	1.36	0.73740
Critical Think	1.27	0.78479
Openness	1.24	0.80381
Conscience	1.24	0.80952
Agree	1.19	0.84066
Extraversion	1.15	0.86727
Neuroticism	1.06	0.94349
Mean VIF	1.22	

Structural Equation Modelling

To estimate the mediating role of resilience on the impact of personality traits on innovation companies, the structural equation modelling (SEM) was used. The result of the of the structural equation model showing the path co-efficient of the mediating role of resilience on the relationship between personality traits dimensions and innovation competences dimensions are reported in Appendix B. The model was equally evaluated with permissible goodness of fit indices reported in Table 6.

Table 6: SEM Goodness of fit

Fit statistic	Value	Description
Likelihood ratio		
p > chi2	0.000	
chi2_bs(810)	11917.5	baseline vs. saturated
p > chi2	0.000	
Population error		
RMSEA	0.063	Root mean squared error of approx.
90% CI, lower bound	0.060	
upper bound	0.066	
pclose	0.000	Probability RMSEA <= 0.05
Information criteria		
AIC	50032.4	Akaike's information criterion
BIC	50640.9	Bayesian information criterion
Baseline comparison		
CFI	0.86	Comparative fit index
TLI	0.85	Tucker–Lewis's index
Size of residuals		
CD	0.993	Coefficient of determination

Results and Findings

The analysis of Creativity, Resilience and Personality Traits

Table 7 presents the structural equation modelling analysis of resilience, personality traits and Innovation competences. As shown in the table, there is strong evidence that resilience is a significant mediator between the effects of personality traits on creativity. Specifically, agreeableness (β=.26, p<0.01) extraversion (β=.35, 1) conscientiousness (β=.65, *p*<0.01) an openness to experience (β=.51, *p*<0.01) are positive and indirect significant predictors of creativity while neuroticism a negative predictor β=-.21, p<0.01). These figures imply that an increase in agreeableness, extraversion conscientiousness and openness to experience by one unit would increase creativity by the predictors' respective beta coefficient through resilience.

In contrast, a one unit increase in neuroticism will indirectly decrease creativity by the beta co-efficient of the latter. Based on this reported result, we confirmed the following hypotheses H1a, H2a, H3a, H4a and H5a

The analysis of Teamwork, Resilience, Personality Traits

The results of the SEM structural model of the indirect effects of personality traits on innovation competences through resilience is shown in the Table 7. Similar to the result on creativity, from the row we can see that all the dimensions of the personality traits have a significant indirect effect on teamwork. All traits except neuroticism, as expected, have a positive relationship with teamwork; the predictors co-efficient are agreeableness (β= .19, $p<0.01$), extraversion (β= .26, $p<0.01$), conscientiousness (β= .49, $p<0.01$) and openness to experience β= .38, $p<0.01$) and neuroticism (β= -16, $p<0.01$). So, we interpret them that through the mechanism of resilience, Teamwork will increase by the respective co-efficient of agreeableness, extraversion, conscientiousness, and openness as they increase by one unit. But Teamwork will decrease by the beta value of neuroticism when the latter increase by one unit. With these results, the following hypotheses H1d, H2d, H3d, H4d and H5d were supported.

The analysis of Initiative, Resilience and Personality Traits

The indirect results of structural model of the impact of personality traits on initiative through resilience can be seen in Table 7. From the row, we can see the following, first that all predictors are significant with extraversion and conscientious having higher co-efficient. Second, neuroticism (β= -.24, $p<0.01$) is an indirect and negative predictor of initiative while agreeableness (β= .28, $p<0.01$) extraversion (β= .38, $p<0.01$) conscientiousness (β= .72, $p<0.01$) and openness to experience (β= .24, $p<0.01$) are positive predictors. A unit increase in the positive predictors will increase initiative (through resilience) by their respective co-efficient whereas a unit increase in neuroticism will decrease initiative by corresponding co-efficient. Putting these together, we confirm the following hypotheses H1b, H2b, H3b, H4b and H5b.

The Analysis of Critical Thinking, Resilience and Personality Traits

Table 7 reports the indirect effect of the structural equation model of personality traits, resilience, and critical thinking. As can be seen from the row, all facets of personality traits are significant and indirect predictors of critical thinking; four of which are positive and one negative. As reported in the table, it is obvious that conscientiousness (β= .39, $p<0.01$) and openness to experience (β= .30, $p<0.01$) have relatively higher predictive co-efficient than

agreeableness (β= .15, p<0.01), extraversion (β= .21, p<0.01) and neuroticism (β= -.13, p<0.01). From the foregoing, it can be implied that a one-unit increase in the positive predictors would increase critical thinking by their respective co-efficient through resilience. In contrast, a unit increase in neuroticism would decrease critical thinking by the co-efficient of the later through resilience. Overall, the results support the following hypotheses H1c, H2c, H3c, H4c and H5c.

The Analysis of Networking, Resilience and Personality Traits

Table 7 provides the result obtained from the model of personality trats as indirect predictors of Networking As the row shows, there is a significant indirect relationship between the five dimensions of personality traits and networking. Except for neuroticism, with a negative relationship, the rest of the trait's factors are positive predictors of networking. Again, as it was with previous results, conscientiousness (β= .63, p<0.01) and openness (β= .49, p<0.01) have higher predictive co-efficient than agreeableness (β= .25, p<0.01), extraversion (β= .33, p<0.01) and neuroticism. (β= -.21, p<0.01). The predicted value of networking would equal to the respective co-efficient of the positive predictors if the later increase by one unit. On the other hand, networking will decrease by the co-efficient value of neuroticism if the later increases by one unit. Hence, from the results support the following hypotheses H1e, H2e, H3e, H4e and H5e.

Taken together, these results as shown in table 7 provides strong evidence that resilience mediates the impact of agreeableness, extraversion conscientiousness, openness to experience and neuroticism on the five dimensions of innovation competences - creativity, critical thinking, initiative, networking, and teamwork. Across all dimensions of innovation competences, the results show the following: first, conscientiousness have the greatest impact followed by openness to experience. Second, all trait factors, except neuroticism are positive predictors of innovation competences.

Table 7: Summary of SEM results

	Innovation Competences									
	Creativity		Critical think.		Teamwork		Networking		Initiative	
Variables mediated by Resilience	Beta	p value	Beta	p value	Beta	Beta	Beta	p value	Beta	p value
Openness	0.51	0.003	0.30	0.000	0.38	0.000	0.49	0.000	0.56	0.000
Agreeable	0.26	0.000	0.15	0.006	0.19	0.003	0.25	0.003	0.28	0.003
Extraversion	0.35	0.000	0.21	0.001	0.26	0.000	0.33	0.000	0.38	0.000
Conscient.	0.65	0.000	0.39	0.000	0.49	0.000	0.63	0.000	0.72	0.000
Neuroticism	-0.21	0.004	-0.13	0.008	-0.16	0.005	-0.21	0.000	-0.24	0.004

Discussion of Findings

This study investigates the role of resilience on the relationship between the big five personality traits on the five dimensions of innovation competences. It was hypothesized that resilience mediates the impact of big five personality traits on creativity, critical thinking, initiative, networking, and teamwork. In the following paragraphs, findings of study results are appropriately discussed.

In this study, openness to experience was found to indirect predict creativity, critical thinking, initiative teamwork and networking thorough resilience. These results corroborate the findings of similar previous studies on personality traits and creativity, and creative abilities (Amin et al., 2020; Hirsh & Peterson, 2008; Kaufman et al., 2016; Raja & Johns, 2010) and critical thinking (Beaty et al., 2016; Feist, 2019; Yao & Li, 2021). One possible explanation could be that creativity which involves the generation of new methods, ideas and solutions may come with so many setbacks and failures before the final breakthrough. Thus, people high on openness are better able to cope with the stress associated with failures because of their rich experience with similar situations. This stress coping ability helps increases the chances of their creative process. Another possible explanation could be that opened people with their high level of imagination and intellect are better able to explore a variety of resources available to support them through their period of adversity. Equipped with knowledge of our people in similar difficult situation survived, opened people become encouraged that would sure sail through their dark days/hours and tough times don't forever. So, we may imply that that the more opened to experience, the higher the level of resilience and thus the higher the level of innovative competences

Another important finding is that resilience also mediates the impact of conscientiousness on creativity, critical thinking, initiative teamwork and networking thorough resilience. This supports evidence from previous studies on personality traits and innovativeness and creativity (Ali, 2019; Amin et al., 2020; Chen, 2016) and initiative (Saatci & Okvaci, 2020b). This can be explained in part that innovation and problem solving would require taking actions to implement new ideas and these actions can become very stressful due to uncertainties and risks. Focus, discipline and attention would therefore become helpful to manage such difficult and stressful endeavors. Highly conscientious people are very focused, disciplined and result oriented. Hence, they can manage these difficulties in the way of innovation and then deliver the much-desired solutions to specific problems. So, we might say

that the more conscientious an individual is, the higher his level of resilience and the higher the level of resilience. the more creative he becomes.

In addition, the result of this study shows that the impact of extraversion on innovation competences is also mediated by resilience. These results are like those obtained in the studies by Amin et al. (2020), Furnham & Bachtiar (2008) and Saatci & Ovaci (2020). Social network: colleagues, collaborators, and friends, can provide valuable support for managing adverse situations and recovering from adversities. The bigger the network, the more support is available to extraverted individuals during their trying times. With larger social network, extraverted people can increase their resilience level. Another perspective to this explanation is that as part of the process of innovation, creative people must be mobilized to execute and operationalize new ideas to achieve the desired solutions, products, or services. Here, again extraversion becomes a vantage to leverage to build great team and receive feedback from others on innovative initiatives. Large social cycle provides support for resilience, and in turn promotes teamwork and networking.

On the question of whether resilience mediates the relationship between agreeableness and innovation competences, this study found that resilience mediates the role between agreeableness and innovation competences. This result goes in line with those of Ali (2019), Batey (2008) and Yao & Li (2021). Though the link between agreeableness and resilience may not be too clear, however one might explain that highly agreeable people can build congenial relationship in harmonious atmosphere and thus get acceptances from others. This acceptance provides the much-needed psychological relief and calmness when handling social conflict arising from collaborative activities.

Expectedly, the mediating role of resilience on the impact of neuroticism on innovation competences is clear. The study confirms that the negative effect of neuroticism on innovation competences is mediated by resilience. We find similar result with (Ali, 2019). The observed relationship between resilience, neuroticism and innovation competences might be explained this way: being resilient requires a lot of emotional stability. Neuroticism personalities are overreactive with so much negative feelings. Thus, this emotional instability might make them less resilient and their therefore reduce their level of teamwork, networking, and creativity.

Conclusion

The main goal of this study was to investigate the impact of resilience on the influence of Big five personality traits on all the five dimensions of innovation competences – creativity, critical thinking, networking, and teamwork. Using structural equation modelling, the results of this investigation show that resilience mediates the effect of big five personality traits on the five dimensions of innovation competences

First, openness to experiences increases the students' level of creativity, critical thinking, networking, initiative, and networking through resilience. This implies that students with high tendency of exploring different interests and activities are more likely to cope with stress and overcome difficulties better. And this would support their level of resilience and then enhance their level of innovation competences.

Second, conscientiousness indirectly and positively predicts creativity, critical thinking, initiatives, networking, and teamwork. The more dutiful and focused the students are the higher their capacity to overcome adversity and then the higher their levels of competences.

Thirdly, people’s level of agreeableness (trustworthiness) sustains heir cordial relationships. This warm relationship would provide them with support them to pull through adversity. And this increase in level of resilience would increase their resilience level during adversity positively impacts their impacts their level of resilience which in turn supports their innovation competences. The fourth finding relates to extraversion the social trait. Likewise, the more of it a student possess, the more resilient and the higher his level innovation competences.

Conversely, neuroticism which negatively and indirectly influences innovation competences. Students with high level of emotional instability are less resilient and eventually displays low level of innovation competences.

Implications

There could be several implications of this study related to organization, universities, and government of the economy.

1. Theoretical implication of this study is that it contributes in the existing theory by determining the role of resilience as a mediator in the innovative competence and personality traits.
2. The second implication of this study can be that it can help the universities to design the various learning strategies and pedagogy methods like problem solving activities in the context of discipline-based curriculum and courses that can develop and enhance the innovative competences and resilience among students.
3. The third implication is for business or employers that they should design training programs for employees that can foster the innovative behaviors and enhance the resilience in the workplace. So, they would be better able to cope with stress and adversity.
4. The fourth implication of this study could be for the government to design the policies that could encourage the institutions and organizations to design the programs that can enhance the innovative behavior in current and future labor force so that they can productively contribute towards economy.
5. This study can help the adolescents, universities, and employers to understand the importance of resilience which can help the individuals to prevent from mental issues like depression and enhance the productivity in long-term.
6. Finally, this study can be helpful for the counselors and psychologist that provides individual or group counseling services to help them to relieve the stress improve well-being among individuals. It will help them by identifying, understand, express, and manage their emotions through various activities and short-trainings and support.

Limitations of the study

Nothing is perfect in the world; everything has positive as well as negative side. Same like this study has few limitations but those limitations are not potential and not affecting the results. These can be overcome in the future research.

1. First limitation of this study is that due to the practical constraints e.g., survey length and respondent's burden, single item measure is used for personality traits. Even single item measure is suggested based on certain conditions, the fact that multi-item measure is better than using single item measure.
2. Second limitation is, all the data was collected on self-reported measures that means individuals rated themselves based on self-understandings, which can cause response bias and increase the chances of low validity and reliability of the scale. However, in this study, reliability and validity was more than standard acceptable value. But for the in-depth insights, responses of the other people on the reaction or the activities of individuals will be considered better such as family members, teachers, and employers etc.
3. Third limitation is that the lack of control variables, linked to participants' demographics, which might have important explanatory effects that are not addressed in this study. They were only used in descriptive studies.

Future Research

Following recommendations are provided for the future work:

1. Longitudinal research can also be work for this type of study.
2. Comparatives approaches can be done based on demographic variables, such as age, gender or work experience.
3. Cultural factors can also be considered as one of the main variables in future study because personality traits and resilience can differ based on culture. Stress and depression vary with nation to nation. It can be because of collectivistic and individualistic society based on the Hofstede model.
4. It is recommended to use multi-items measure for the any variables that are mentioned in this study.
5. Based on the limitation, it is also recommended to collect data that is not self-reported.

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Appendix A

Questionnaire used in the study

The study used a survey instrument which is made up of set of questions which are meant to elicit responses to measure the three (3) constructs innovation competences items, personality traits and resilience

Innovation competences items

Please, evaluate from 1 (absolutely disagree) to 5 (absolutely agree) your ability to...

1. think differently and adopt different perspectives
2. be attentive when others are speaking, and respond effectively to others' comments during the conversation
3. use intuition and own knowledge to start actions
4. invite feedback and comments
5. foster improvements in work organization
6. obtain constructive comments from colleagues
7. find new ways to implement ideas
8. identify sources of conflict between oneself and others, or among other people, and to take steps to overcome disharmony
9. take an acceptable level of risk to support new ideas
10. go beyond expectations in the assignment, task, or job description without being asked
11. meet people with different kinds of ideas and perspectives to extend your own knowledge domain
12. convince people to support an innovative idea
13. systematically introduce new ideas into work practices
14. act quickly and energetically
15. generate original solutions for problems or to opportunities
16. use trial and error for problem solving
17. develop and experiment with new ways of problem solving
18. acquire, assimilate, transform, and exploit external knowledge to establish, manage and learn from informal organizational ties
19. challenge the status quo
20. face the task from different points of view
21. make suggestions to improve current process, products, or services
22. present novel ideas
23. forecast impact on users
24. show inventiveness in using resources
25. search out new working methods, techniques, or instruments
26. provide constructive feedback, cooperation, coaching or help to team colleagues
27. work well with others, understanding their needs and being sympathetic with them

28. share timely information with the appropriate stakeholders
29. consult about essential changes
30. build relationships outside the team/organization
31. refine ideas into a useful form
32. engage outsiders of the core work group from the beginning
33. ask “Why?” and “Why not?” and “What if?” with a purpose
34. work in multidisciplinary environments

Personality Traits

Here are several characteristics that may or may not apply to you. For example, do you agree that you are someone who likes to spend time with others? Please, evaluate from 1 (absolutely disagree) to 5 (absolutely agree) indicate the extent to which you agree or disagree with that statement.

I see myself as someone who

1. Is reserved
2. Is generally trusting
3. tends to be lazy
4. is relaxed, handles stress well
5. has few artistic interests
6. is outgoing, sociable
7. tends to find fault with others
8. does a thorough job
9. gets nervous easily
10. has an active imagination
11. makes plans and follows through with them

Resilience items

1. I tend to bounce back quickly after hard times.
2. It does not take me long to recover from a stressful event.
3. I tend to take a long time to get over setbacks in my life.

Appendix B

Structural Equation Model: Indirect

Indirect Effects

	OIM					
	Coefficient	**std. err.**	**z**	**P>|z|**	**[95% conf.**	**interval]**
Creativity						
Resilience	0	(no path)				
Agreeableness	0.2626422	0.0881551	2.98	0.003	0.0898614	0.435423
Extraversion	0.3520673	0.0977674	3.6	0.000	0.1604467	0.543688
Conscientiousness	0.6592149	0.1286747	5.12	0.000	0.4070172	0.9114127
Openness to experience	0.5144927	0.1222882	4.21	0.000	0.2748123	0.7541731
Neuroticism	-0.2189205	0.0766577	-2.86	0.004	-0.369167	-0.0686741
Teamwork						
Resilience	0	(no path)				
Agreeableness	0.1987278	0.0673299	2.95	0.003	0.0667636	0.3306921
Extraversion	0.2663913	0.0749593	3.55	0.000	0.1194737	0.4133089
Conscientiousness	0.498794	0.0999863	4.99	0.000	0.3028245	0.6947636
Openness to experience	0.3892901	0.0942024	4.13	0.000	0.2046569	0.5739234
Neuroticism	-0.1656459	0.0584972	-2.83	0.005	-0.2802984	-0.0509935
Initiatives						
Resilience	0	(no path)				
Agreeableness	0.288187	0.0965271	2.99	0.003	0.0989975	0.4773766
Extraversion	0.3863098	0.1069275	3.61	0.000	0.1767358	0.5958838
Conscientiousness	0.7233308	0.1402627	5.16	0.000	0.4484211	0.9982406
Openness to experience	0.5645328	0.1335641	4.23	0.000	0.3027519	0.8263136
Neuroticism	-0.240213	0.0839501	-2.86	0.004	-0.4047521	-0.0756739
Critical Thinking						
Resilience	0	(no path)				
Agreeableness	0.156964	0.0567128	2.77	0.006	0.0458089	0.2681191
Extraversion	0.2104076	0.0648373	3.25	0.001	0.0833289	0.3374863
Conscientiousness	0.3939695	0.0932441	4.23	0.000	0.2112144	0.5767247
Openness to experience	0.3074785	0.0838627	3.67	0.000	0.1431106	0.4718465
Neuroticism	-0.1308345	0.0490461	-2.67	0.008	-0.226963	-0.034706

Networking						
Resilience	0	(no path)				
Agreeableness	0.2524821	0.0849918	2.97	0.003	0.0859012	0.419063
Extraversion	0.3384479	0.0943457	3.59	0.000	0.1535338	0.5233621
Conscientiousness	0.6337138	0.1246649	5.08	0.000	0.389375	0.8780525
Openness to experience	0.49459	0.1181617	4.19	0.000	0.2629973	0.7261827
Neuroticism	-0.2104518	0.073874	-2.85	0.004	-0.3552422	-0.0656614

Resilience		
Agreeableness	0	(no path)
Extraversion	0	(no path)
Conscientiousness	0	(no path)
Openness to experience	0	(no path)
Neuroticism	0	(no path)

Appendix C

Structural Equation Model: Total effects

	OIM					
	Coefficient	**std. err.**	**z**	**P>|z|**	**[95% conf.**	**interval]**
Structural Model						
Creativity						
Resilience	2.0204	0.3329	6.07	0.000	1.3679	2.6730
Agreeableness	0.1423	0.0278	5.12	0.000	0.0879	0.1967
Extraversion	0.1403	0.0280	5.02	0.000	0.0855	0.1951
Conscientiousness	0.1885	0.0279	6.76	0.000	0.1338	0.2431
Openness to experience	0.2375	0.0336	7.07	0.000	0.1716	0.3033
Neuroticism	-0.0574	0.0237	-2.42	0.016	-0.1039	-0.0109
Teamwork						
Resilience	1.5288	0.2613	5.8500	0.0000	1.0165	2.0410
Agreeableness	0.1525	0.0274	5.5600	0.0000	0.0987	0.2062
Extraversion	0.1153	0.0272	4.2400	0.0000	0.0621	0.1686
Conscientiousness	0.2300	0.0286	8.0500	0.0000	0.1740	0.2860
Openness to experience	0.1873	0.0323	5.7900	0.0000	0.1239	0.2507
Neuroticism	-0.0472	0.0231	-2.0400	0.0410	-0.0925	-0.0019
Initiatives						
Resilience	2.2170	0.3619	6.1300	0.0000	1.5076	2.9264
Agreeableness	0.1535	0.0331	4.6400	0.0000	0.0887	0.2183
Extraversion	0.1827	0.0337	5.4200	0.0000	0.1167	0.2488
Conscientiousness	0.2510	0.0340	7.3700	0.0000	0.1843	0.3177
Openness to experience	0.2520	0.0395	6.3900	0.0000	0.1747	0.3294
Neuroticism	-0.0600	0.0288	-2.0900	0.0370	-0.1164	-0.0037
Critical Thinking						
Resilience	1.2075	0.2562	4.71000	0.0000	0.7053	1.7097
Agreeableness	0.0893	0.0216	4.14000	0.0000	0.0470	0.1316
Extraversion	0.0617	0.0199	3.10000	0.0020	0.0227	0.1006
Conscientiousness	0.0943	0.0216	4.38000	0.0000	0.0521	0.1365
Openness to experience	0.1195	0.0268	4.47000	0.0000	0.0671	0.1720
Neuroticism	-0.0261	0.0158	-1.65000	0.0990	-0.0572	0.0049

Networking						
Resilience	1.9423	0.3235	6	0.000	1.3082	2.5764
Agreeableness	0.1676	0.0308	5.45	0.000	0.1073	0.2278
Extraversion	0.1453	0.0313	4.64	0.000	0.0839	0.2067
Conscientiousness	0.2154	0.0308	6.99	0.000	0.1550	0.2758
Openness to experience	0.1914	0.0361	5.3	0.000	0.1206	0.2621
Neuroticism	-0.0607	0.0264	-2.3	0.021	-0.1124	-0.0090
Resilience						
Agreeableness	0.1299924	0.0394597	3.29	0.001	0.0527	0.20733
Extraversion	0.1742525	0.040035	4.35	0.000	0.0958	0.25272
Conscientiousness	0.3262724	0.0424586	7.68	0.000	0.2431	0.40949
Openness to experience	0.2546434	0.0489338	5.2	0.000	0.1587	0.35055
Neuroticism	-0.1083527	0.0342335	-3.17	0.002	-0.175	-0.04126